

Two-Step Equations

Name
Date
Course

$$\textcircled{1} \quad 7x - 4 = 3$$

$$4 + 7x - 4 = 3 + 4$$

$$7x = 7$$

$$\frac{7x}{7} = \frac{7}{7}$$

$$\boxed{x = 1}$$

$$\textcircled{3} \quad -6z - 8 = -1$$

$$8 - 6z - 8 = -1 + 8$$

$$-6z = 7$$

$$\frac{-6z}{-6} = \frac{7}{-6}$$

$$\boxed{z = -\frac{7}{6}}$$

$$\textcircled{5} \quad -3 - 7y = -10$$

$$3 - 3 - 7y = -10 + 3$$

$$-7y = -7$$

$$\frac{-7y}{-7} = \frac{-7}{-7}$$

$$\boxed{y = 1}$$

$$\textcircled{2} \quad 3 = 9 + \frac{y}{5}$$

$$-9 + 3 = 9 + \frac{y}{5} - 9$$

$$-6 = \frac{y}{5}$$

$$(5)(-6) = \frac{y}{5}(5)$$

$$\boxed{-30 = y}$$

$$\textcircled{4} \quad -5 = \frac{x}{-3} + 4$$

$$-4 - 5 = \frac{x}{-3} + 4 - 4$$

$$-9 = \frac{x}{-3}$$

$$(-3)(-9) = \frac{x}{-3}(-3)$$

$$\boxed{27 = x}$$

$$\textcircled{6} \quad 8 = 6x + 6$$

$$-6 + 8 = 6x + 6 - 6$$

$$2 = 6x$$

$$\frac{2}{6} = \frac{6x}{6}$$

$$\boxed{\frac{1}{3} = x}$$

$$\textcircled{7} \quad \frac{y}{5} + 7 = -2$$

$$-7 + \frac{y}{5} + 7 = -2 - 7$$

$$\frac{y}{5} = -9$$

$$(5)\frac{y}{5} = -9(5)$$

$$\boxed{y = -45}$$

$$\textcircled{9} \quad 5 = 7 - 2x$$

$$-7 + 5 = 7 - 2x - 7$$

$$-2 = -2x$$

$$\frac{-2}{-2} = \frac{-2x}{-2}$$

$$\boxed{1 = x}$$

$$\textcircled{11} \quad \frac{x}{4} + 9 = 13$$

$$-9 + \frac{x}{4} + 9 = 13 - 9$$

$$\frac{x}{4} = 4$$

$$(4)\frac{x}{4} = 4(4)$$

$$\boxed{x = 16}$$

$$\textcircled{8} \quad -3z + 5 = 9$$

$$-5 - 3z + 5 = 9 - 5$$

$$-3z = 4$$

$$\frac{-3z}{-3} = \frac{4}{-3}$$

$$\boxed{z = -\frac{4}{3}}$$

$$\textcircled{10} \quad 10 = -8x - 1$$

$$1 + 10 = -8x - 1 + 1$$

$$11 = -8x$$

$$\frac{11}{-8} = \frac{-8x}{-8}$$

$$\boxed{-\frac{11}{8} = x}$$

$$\textcircled{12} \quad -7 = 5z - 2$$

$$2 - 7 = 5z - 2 + 2$$

$$\frac{-5}{5} = \frac{5z}{5}$$

$$\boxed{-1 = z}$$

$$\begin{aligned}
 (13) \quad & -2x + 3 = -1 \\
 & -3 - 2x + 3 = -1 - 3 \\
 & -2x = -4 \\
 & \frac{-2x}{-2} = \frac{-4}{-2} \\
 & \boxed{x = 2}
 \end{aligned}$$

$$\begin{aligned}
 (14) \quad & -9 = -2 + \frac{y}{4} \\
 & 2 - 9 = -2 + \frac{y}{4} + 2 \\
 & -7 = \frac{y}{4} \\
 & (4)(-7) = \frac{y}{4}(4) \\
 & \boxed{-28 = y}
 \end{aligned}$$

$$\begin{aligned}
 (15) \quad & 11 - 3z = 4 \\
 & -11 + 11 - 3z = 4 - 11 \\
 & -3z = -7 \\
 & \frac{-3z}{-3} = \frac{-7}{-3} \\
 & \boxed{z = \frac{7}{3}}
 \end{aligned}$$

$$\begin{aligned}
 (16) \quad & \frac{x}{-6} + 10 = -2 \\
 & -10 + \frac{x}{-6} + 10 = -2 - 10 \\
 & \frac{x}{-6} = -12 \\
 & (-6)\frac{x}{-6} = -12(-6) \\
 & \boxed{x = 72}
 \end{aligned}$$

$$\begin{aligned}
 (17) \quad & -5x - 4 = -7 \\
 & 4 - 5x - 4 = -7 + 4 \\
 & -5x = -3 \\
 & \frac{-5x}{-5} = \frac{-3}{-5} \\
 & \boxed{x = \frac{3}{5}}
 \end{aligned}$$

$$\begin{aligned}
 (18) \quad & -7 - 10y = 4 \\
 & 7 - 7 - 10y = 4 + 7 \\
 & -10y = 11 \\
 & \frac{-10y}{-10} = \frac{11}{-10} \\
 & \boxed{y = -\frac{11}{10}}
 \end{aligned}$$

$$\begin{aligned}
 (19) \quad & 10 = -\frac{y}{5} + 6 \\
 & -6 + 10 = -\frac{y}{5} + 6 - 6 \\
 & (-5)4 = -\frac{y}{5}(-5) \\
 & \boxed{-20 = y}
 \end{aligned}$$

$$\begin{aligned}
 (21) \quad & -1 = 4 - \frac{z}{2} \\
 & -4 - 1 = 4 - \frac{z}{2} - 4 \\
 & (-2)(-5) = -\frac{z}{2}(-2) \\
 & \boxed{10 = z}
 \end{aligned}$$

$$\begin{aligned}
 (23) \quad & -x + 7 = 9 \\
 & -7 - x + 7 = 9 - 7 \\
 & -x = 2 \\
 & \frac{-1x}{-1} = \frac{2}{-1} \\
 & \boxed{x = -2}
 \end{aligned}$$

$$\begin{aligned}
 (20) \quad & -\frac{x}{2} - 6 = 5 \\
 & 6 - \frac{x}{2} - 6 = 5 + 6 \\
 & (-2)\left(-\frac{x}{2}\right) = 11(-2) \\
 & \boxed{x = -22}
 \end{aligned}$$

$$\begin{aligned}
 (22) \quad & -3 - \frac{x}{8} = -7 \\
 & 3 - 3 - \frac{x}{8} = -7 + 3 \\
 & (-8)\frac{x}{8} = -4(-8) \\
 & \boxed{x = 32}
 \end{aligned}$$

$$\begin{aligned}
 (24) \quad & 12 = -y + 4 \\
 & -4 + 12 = -y + 4 - 4 \\
 & 8 = -y \\
 & \frac{8}{-1} = \frac{-1y}{-1} \\
 & \boxed{-8 = y}
 \end{aligned}$$

$$\begin{aligned} \textcircled{25} \quad 9 - x &= 6 \\ -9 + 9 - x &= 6 - 9 \\ -x &= -3 \\ \frac{-1x}{-1} &= \frac{-3}{-1} \\ \boxed{x = 3} \end{aligned}$$

$$\begin{aligned} \textcircled{26} \quad 4 &= -10 - y \\ 10 + 4 &= -10 - y + 10 \\ 14 &= -y \\ \frac{14}{-1} &= \frac{-y}{-1} \\ \boxed{-14 = y} \end{aligned}$$

$$\begin{aligned} \textcircled{27} \quad 7 &= 4x + 7 \\ -7 + 7 &= 4x + 7 - 7 \\ 0 &= 4x \\ \frac{0}{4} &= \frac{4x}{4} \\ \boxed{0 = x} \end{aligned}$$

$$\begin{aligned} \textcircled{28} \quad -3z + 6 &= 6 \\ -6 - 3z + 6 &= 6 - 6 \\ -3z &= 0 \\ \frac{-3z}{-3} &= \frac{0}{-3} \\ \boxed{z = 0} \end{aligned}$$

$$\begin{aligned} \textcircled{29} \quad \frac{y}{-5} - 1 &= -1 \\ 1 + \frac{y}{-5} - 1 &= -1 + 1 \\ \frac{y}{-5} &= 0 \\ (-5) \frac{y}{-5} &= 0(-5) \\ \boxed{y = 0} \end{aligned}$$

$$\begin{aligned} \textcircled{30} \quad -8 &= \frac{z}{2} - 8 \\ 8 - 8 &= \frac{z}{2} - 8 + 8 \\ 0 &= \frac{z}{2} \\ (2) 0 &= \frac{z}{2} (2) \\ \boxed{0 = z} \end{aligned}$$